

Manual Launch Guide — S1 Handover VoLTE Lab

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Manual Launch Guide — S1 Handover VoLTE Lab

Complete step-by-step procedure to launch the lab manually, from cold start to live Wireshark capture during handover.

Duration: ~10 minutes

Prerequisites: All hardware connected, SIM cards provisioned

Quick Reference

Component	Machine	Command
Core + IMS	PC1	<code>docker-compose up -d</code>
eNB1	PC1	<code>sudo srseNB /tmp/enb1_handover.conf</code>
eNB2	PC2	<code>sudo srseNB /tmp/enb2_handover.conf</code>
Wireshark	PC1	<code>wireshark -i any -k -f "sctp or gtp"</code>

Phase 1: Pre-flight Checks (2 min)

1.1 Verify Network Connectivity

```
# On PC1
ping -c 2 192.168.1.101    # PC2 reachable
ping -c 2 192.168.1.102    # MME IP (may fail if not started yet)
```

1.2 Check bladeRF Devices

```
# On PC1
bladeRF-cli -p
# Expected: Serial XXXXXXXXXX, bladeRF 2.0 (bladeRF xA4)

# On PC2 (via SSH)
ssh f2g@192.168.1.101 "bladeRF-cli -p"
```

1.3 Verify Config Files Exist

```
# On PC1
ls -la /tmp/enb1_handover.conf /tmp/rr_enb1_ho.conf

# On PC2
ssh f2g@192.168.1.101 "ls -la /tmp/enb2_handover.conf /tmp/rr_enb2_ho.conf"
```

If config files are missing, copy them:

```
# PC1
cp ~/Desktop/Handover/configs/enb1/enb1_handover.conf /tmp/
cp ~/Desktop/Handover/configs/enb1/rr_enb1_ho.conf /tmp/

# PC2
scp ~/Desktop/Handover/configs/enb2/enb2_handover.conf f2g@192.168.1.101:/tmp/
scp ~/Desktop/Handover/configs/enb2/rr_enb2_ho.conf f2g@192.168.1.101:/tmp/
```

Phase 2: Start Core Network (2 min)

2.1 Start Docker Containers

```
cd /home/f2g/Telecom/Core-Network/openimss
docker-compose up -d
```

Wait 10 seconds for all services to initialize.

2.2 Verify Core Services

```
# Check containers running
docker ps --format "table {{.Names}}\t{{.Status}}" | head -15
```

```
# Check MME is listening
docker logs mme 2>&1 | tail -5
# Look for: "slap_server() [INFO]: Waiting for eNB connection"

# Check IMS Diameter connections
docker logs pcrf 2>&1 | grep -i "CONNECTED"
# Look for: "CONNECTED TO 'pcscf'"
```

2.3 Create macvlan Network (First Time Only)

```
# Create macvlan for MME external access
docker network create -d macvlan \
  --subnet=192.168.1.0/24 \
  --gateway=192.168.1.1 \
  -o parent=enp3s0 \
  macvlan_net

# Connect MME to macvlan
docker network connect --ip 192.168.1.102 macvlan_net mme
```

2.4 Verify MME External Connectivity

```
# From PC2, test S1 connectivity to MME
ssh f2g@192.168.1.101 "nc -zv 192.168.1.102 36412"
# Expected: Connection to 192.168.1.102 36412 port [tcp/*] succeeded!
```

Phase 3: Start Wireshark Live Capture (1 min)

3.1 Launch Wireshark with S1AP/GTP Filter

Open a new terminal on PC1:

```
# Option A: Capture all interfaces with SCTP and GTP
sudo wireshark -i any -k -f "sctp or udp port 2152"

# Option B: Capture only macvlan interface
sudo wireshark -i enp3s0 -k -f "sctp or udp port 2152"

# Option C: CLI capture with tshark
sudo tshark -i any -f "sctp or udp port 2152" -Y "slap or gtpv2"
```

3.2 Wireshark Display Filter for Handover

In Wireshark filter bar, apply:

```
slap.procedureCode == 0 || slap.procedureCode == 1 || slap.procedureCode == 5
```

Or for all S1AP messages:

```
slap
```

3.3 Key S1AP Messages to Watch

Procedure Code	Message	Direction
0	HandoverPreparation	eNB → MME
1	HandoverResourceAllocation	MME → eNB
5	HandoverNotification	eNB → MME
3	PathSwitchRequest	eNB → MME

Phase 4: Start eNB1 on PC1 (1 min)

4.1 Launch srsENB on PC1

```
# Terminal 1 on PC1
cd /tmp
sudo srsenb enb1_handover.conf
```

4.2 Expected Output

```
--- Software Radio Systems LTE eNodeB ---

Opening 1 channels in RF device=bladeRF with args=
Set RX antenna to RX
...
==== eNodeB started ===
Type <t> to view trace
Setting frequency: DL=2140.0 MHz, UL=1950.0 MHz
Found bladeRF serial XXXXXXXXXX
```

4.3 Verify S1 Connection in MME Logs

```
# In another terminal
docker logs -f mme 2>&1 | grep -i "enb\|s1ap"
# Look for: "eNB-S1 accepted[192.168.1.100]"
# Look for: "Number of eNBs is now 1"
```

Phase 5: Start eNB2 on PC2 (1 min)

5.1 SSH to PC2 and Launch

```
# From PC1, SSH to PC2
ssh f2g@192.168.1.101

# On PC2
cd /tmp
sudo srsenb enb2_handover.conf
```

Password: 1234

5.2 Alternative: Launch from PC1 via SSH

```
# One-liner from PC1
sshpass -p "1234" ssh f2g@192.168.1.101 "cd /tmp && sudo -S srsenb enb2_handover.conf" <<< "1234"
```

5.3 Verify Both eNBs Connected

```
docker logs mme 2>&1 | grep "Number of eNBs"
# Expected: "Number of eNBs is now 2"
```

In Wireshark, you should see: - 2x S1SetupRequest (one from each eNB) - 2x S1SetupResponse (from MME)

Phase 6: Attach UEs and Register IMS (2 min)

6.1 Power On Phones

1. Insert SIM cards (IMSI 001010000123451 and 001010000099901)
2. Power on both phones
3. Wait for LTE attachment (signal bars appear)

6.2 Verify LTE Attachment

On eNB1 terminal, press **t** for trace:

```
-----DL-----UL-----
rnti  cqi  ri  mcs  brate  ok  nok  (%)  pusch  pucch  phr  mcs  brate
  46   15   1   28   5.2M  100   2   2%   -8.0   -4.5   20   24   2.1M
```

6.3 Verify IMS Registration

```
docker logs scscf 2>&1 | grep -i "REGISTER"
# Look for: "REGISTER sip:ims.mnc001.mcc001.3gppnetwork.org"
# Look for: "200 OK" response
```

6.4 Check VoLTE Status on Phones

- Android: Settings → About Phone → SIM Status → “Voice over LTE: Available”
- Or dial *##*4636*##* → Phone Information → VoLTE Provisioned: true

Phase 7: Initiate VoLTE Call (1 min)

7.1 Start Call Monitoring

```
# Terminal on PC1 - watch SIP INVITE
docker logs -f scscf 2>&1 | grep -iE "INVITE|180|200|ACK|BYE"
```

7.2 Make Call

From Phone 1 (1001), dial Phone 2 (1099):

1099

7.3 Verify Call Established

In S-CSCF logs:

```
INVITE sip:1099@ims.mnc001.mcc001.3gppnetwork.org
180 Ringing
200 OK
ACK
```

In Wireshark, verify GTP-U traffic (UDP port 2152) flowing.

Phase 8: Trigger Handover (1 min)

8.1 Move Phone Physically

With call active, move Phone 1 closer to eNB2 and away from eNB1.

Handover triggers when: - RSRP from eNB2 > RSRP from eNB1 + 6 dB (a3_offset) - Condition holds for 480 ms (time_to_trigger)

8.2 Monitor Handover in eNB1

In eNB1 terminal (press t):

```
H0: Starting S1 Handover to eci=0x19C01 (eNB2)
H0: Sending HandoverRequired
H0: Received HandoverCommand
H0: RRC Connection Release (handover complete to target)
```

8.3 Monitor Handover in MME Logs

```
docker logs -f mme 2>&1 | grep -iE "handover|source|target"
```

Expected sequence:

```
[S1AP] Received HandoverRequired from eNB 0x19B
[S1AP] Sending HandoverRequest to eNB 0x19C
[S1AP] Received HandoverRequestAcknowledge from eNB 0x19C
[S1AP] Sending HandoverCommand to eNB 0x19B
[S1AP] Received MMEStatusTransfer
[S1AP] Sending MMEStatusTransfer to target eNB
[S1AP] Received HandoverNotify from eNB 0x19C
[S1AP] Sending UEContextReleaseCommand to source eNB
```

8.4 Verify in Wireshark

Complete S1AP handover sequence:

#	Message	Source	Destination
1	HandoverRequired	eNB1	MME
2	HandoverRequest	MME	eNB2
3	HandoverRequestAcknowledge	eNB2	MME
4	HandoverCommand	MME	eNB1
5	eNBStatusTransfer	eNB1	MME
6	MMEStatusTransfer	MME	eNB2
7	HandoverNotify	eNB2	MME
8	UEContextReleaseCommand	MME	eNB1
9	UEContextReleaseComplete	eNB1	MME

Phase 9: Verify Call Continuity

9.1 Check Audio Still Working

After handover, verify: - No audio interruption - Call still connected - Both parties can hear each other

9.2 Check CellID Changed

On Phone (Android):

####4636#### → Phone Information → Cell ID

- Before HO: CellID = 0x19B01 (eNB1)
- After HO: CellID = 0x19C01 (eNB2)

9.3 Save Wireshark Capture

File → Save As → handover_volte_YYYYMMDD_HHMM.pcapng

Troubleshooting Quick Reference

eNB won't connect to MME

```
# Check MME is reachable
nc -zv 192.168.1.102 36412

# Check MME container
docker logs mme | tail -20

# Restart MME
docker restart mme
```

UE won't attach

```
# Check HSS subscriber
docker exec -it hss /bin/bash
open5gs-dbctl show
```

```
# Verify PLMN on phone matches 001-01
```

IMS registration fails

```
# Check P-CSCF
docker logs pcsf | grep -i error

# Check Diameter Rx
docker logs pcrf | grep -i pcsf

# Restart IMS stack
docker restart pcrf pcsf icscf scscf hss
```

Handover not triggering

```
# Check A3 parameters in rr config
grep -A10 "meas_report_desc" /tmp/rr_enb1_ho.conf

# Verify both cells in meas_cell_list
grep -A10 "meas_cell_list" /tmp/rr_enb1_ho.conf

# Increase logging
sudo srsenb enb1_handover.conf --log.all_level=debug
```

Call drops during handover

```
# Check GTP tunnel
docker logs sgwu | grep -i error

# Check bearer modification
docker logs mme | grep -i "bearer\|modify"

# Verify TAC match (CRITICAL)
grep "tac" /tmp/enb1_handover.conf
ssh f2g@192.168.1.101 "grep tac /tmp/enb2_handover.conf"
# Both must be: tac = 0x0001
```

Complete Startup Script

Save as `start_lab.sh`:

```
#!/bin/bash
# Complete lab startup script

echo "=== Starting VoLTE Handover Lab ==="

# 1. Start Core
```

```

cd /home/f2g/Telecom/Core-Network/openimss
docker-compose up -d
sleep 10

# 2. Verify Core
echo ""
echo "=== Core Status ==="
docker ps --format "table {{.Names}}\t{{.Status}}" | grep -E "mme|hss|cscf"

# 3. Start Wireshark in background
echo ""
echo "=== Starting Wireshark ==="
sudo wireshark -i any -k -f "sctp or udp port 2152" &
sleep 2

# 4. Start eNB1
echo ""
echo "=== Starting eNB1 ==="
gnome-terminal --title="eNB1" -- bash -c "sudo srsenb /tmp/enb1_handover.conf; exec bash"

# 5. Start eNB2 via SSH
echo ""
echo "=== Starting eNB2 on PC2 ==="
gnome-terminal --title="eNB2 (PC2)" -- bash -c "sshpass -p '1234' ssh -t f2g@192.168.1.101 'sudo

echo ""
echo "=== Lab Started ==="
echo "1. Wait for 'Number of eNBs is now 2' in MME logs"
echo "2. Power on phones"
echo "3. Make VoLTE call"
echo "4. Move phone to trigger handover"

```

Shutdown Procedure

```

# 1. Stop eNBs (Ctrl+C in each terminal)

# 2. Stop Core
cd /home/f2g/Telecom/Core-Network/openimss
docker-compose down

# 3. Save Wireshark capture before closing

```

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